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DS 120 Pop Quiz 2

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Instructions

- Unless otherwise instructed by a Teaching Fellow, the time limit for this quiz is **10 minutes**. Please manage your time accordingly.
- At the end of the quiz, please **submit** your completed paper to your Teaching Fellow.
- Only tools required for writing your answers are allowed on the desk. All other items must be put away.

Multiple Choice

Select all that apply; no partial credit

Question 1. Which of the following statements are true about the principle of mathematical induction? (5 points)

- ☒ It is used to prove statements about all natural numbers.
- ☒ It requires proving a base case and an inductive step.
- ☐ It can only be used for statements involving inequalities.
- ☒ If both the base case and the inductive step are true, the statement holds for all $n \in \mathbb{N}$.

Question 2. What is the purpose of the *base case* in an induction proof? (5 points)

- ☒ To verify that the statement holds for the smallest natural number in the domain.
- ☒ To provide a starting point for the inductive chain.
- ☐ To check whether the statement is true for infinitely many values.
- ☐ To skip over trivial cases.

Prove the following using induction:

Show that $4^n - 1$ is divisible by 3 for all $n \in \mathbb{N}$. (10 points)

$$F(n) : 4^n - 1 \% 3 == 0$$

$$F(1) : 4^1 - 1 \% 3 == 0 : 3 \% 3 == 0 \Rightarrow \text{True} \quad \left. \vphantom{F(1)} \right\} \text{Base case holds}$$
$$: 4 - 1 \% 3 == 0$$

Suppose $F(n)$ is true for some k where $k \in \mathbb{N}$

$$F(k+1) : 4^{k+1} - 1 \% 3 == 0$$